Kinds of collisions

antocciphysics | TEACHER KEY | CPA

Learning target: Separate momentum conservation from kinetic-energy conservation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 3; 3 course days later.*
$J=30$ N·s right; $v_f=10$ m/s right.

02. Explain a worked example (worked)
Two separate carts before; joined pair after at 2 m/s in the original direction.

03. Look and explain (model)
Joined carts share final velocity.

04. Build the solution (practice)
2 m/s in original moving cart’s direction.

05. Use the idea (reason)
Kinetic energy decreases but total energy is conserved; deformation, heat and sound may gain energy.

06. Choose a method (mixed) *Review: Unit 2, Day 10; 12 course days later.*
35 m from the base in the launch direction.

07. Exit check (exit)
Sticking; total momentum.

08. STRETCH: Apply it another way (stretch)
Yes, both totals 4 kg·m/s. Additional information about the interaction/energy is needed.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 3; 3 course days later.*
J=7.5 N·s; Δv=12.5 m/s in the force direction.

02. Explain a worked example (worked)
Two separate carts before; joined pair after at 2 m/s in the original direction.

03. Represent and interpret (model)
Joined carts share final velocity.

04. Solve and justify (practice)
Before 9 J; after 6 J; 3 J transformed, one third.

05. Analyze the reasoning (reason)
Kinetic energy decreases but total energy is conserved; deformation, heat and sound may gain energy.

06. Choose a method (mixed) *Review: Unit 2, Day 10; 12 course days later.*
t=3 s; range 18 m; vertical velocity 30 m/s downward.

07. Exit check (exit)
Initially moving mass stops; other moves atu, for ideal 1 D equal-mass elastic collision.

08. STRETCH: Apply it another way (stretch)
All give total 4 kg·m/s for equal masses. Only (2, 2) is a shared sticking velocity; momentum alone does not classify all interactions.